MERN & Next.js

Full Stack Web Development — Course Syllabus

Build enterprise web applications with React 19, Next.js 15, Node.js, TypeScript, PostgreSQL, and AWS deployment.

Core Technologies Mastered

React · Next.js · Node.js · TypeScript

Program Overview

This 16-week, project-driven curriculum takes engineers from modern JavaScript fundamentals to production-level, enterprise-grade full-stack engineering proficiency. Each module combines hands-on coding, architecture review, and real-world project work culminating in a capstone deployment on AWS.

#	Module	Duration	Focus Area
1	Frontend Architecture & Modern JavaScript	Week 1 – 2	Frontend
2	React 19 & State Management	Week 3 – 5	React
3	Next.js 15 & Server Components	Week 6 – 8	Next.js
4	Backend & Database Systems	Week 9 – 11	Backend
5	Security & Authentication	Week 12 – 13	Security
6	DevOps, Docker & AWS	Week 14 – 16	DevOps

Module 1: Frontend Architecture & Modern JavaScript

Duration	Week 1 – 2
Outcome	Ability to structure scalable frontend codebases using modern ES2023+ JavaScript and modular architecture patterns.

Topics Covered

- ES6+ deep dive: destructuring, spread/rest, optional chaining, nullish coalescing
- Asynchronous JavaScript: Promises, async/await, event loop internals
- Module systems: ES Modules vs CommonJS, tree-shaking, code splitting
- Frontend project architecture: feature-based vs layer-based folder structures
- Design patterns: Singleton, Factory, Observer, and Module patterns in JS
- Build tooling: Vite, Webpack fundamentals, Babel, and bundler configuration
- Browser fundamentals: DOM APIs, event delegation, Web Storage, Fetch API
- Package management with npm/yarn/pnpm and monorepo basics (Turborepo/Nx)

Module 2: React 19 & State Management

Duration	Week 3 – 5
Outcome	Production-ready proficiency building complex, performant React applications with modern state management.

Topics Covered

- React 19 fundamentals: components, JSX, props, and the new compiler
- Hooks mastery: useState, useEffect, useMemo, useCallback, useRef, useTransition
- React 19 features: Actions, useActionState, useOptimistic, and the 'use' hook
- Custom hooks and reusable logic composition
- State management strategies: Context API, Zustand, Redux Toolkit
- Form handling and validation with React Hook Form + Zod
- Component design patterns: compound components, render props, HOCs
- Performance optimization: memoization, virtualization, React DevTools profiling
- Testing: Jest, React Testing Library, and component test strategy

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Module 3: Next.js 15 & Server Components

Duration	Week 6 – 8
Outcome	Capability to architect full-stack Next.js applications leveraging the App Router and server-first rendering.

Topics Covered

- App Router architecture: file-based routing, layouts, and route groups
- React Server Components (RSC) vs Client Components: boundaries and data flow
- Data fetching patterns: server actions, streaming, and Suspense
- Rendering strategies: SSR, SSG, ISR, and Partial Prerendering (PPR)
- Server Actions for mutations and progressive enhancement
- Middleware, route handlers, and API routes in Next.js 15
- Caching model: fetch cache, Data Cache, Full Route Cache, Router Cache
- Metadata API, SEO optimization, and Open Graph image generation
- Image, font, and performance optimization with next/image and next/font

Module 4: Backend & Database Systems

Duration	Week 9 – 11
Outcome	Skill to design and build robust, scalable REST/GraphQL APIs backed by relational and NoSQL data stores.

Topics Covered

- Node.js runtime internals: event loop, streams, and worker threads
- Building REST APIs with Express.js / Fastify and layered architecture
- GraphQL fundamentals with Apollo Server and schema design
- PostgreSQL: schema design, indexing, joins, transactions, and normalization
- ORM/query builders: Prisma and Drizzle ORM for type-safe database access
- MongoDB and Mongoose for document-based data modeling (MERN core)
- Caching layers with Redis for sessions, rate limiting, and query caching
- API design best practices: versioning, pagination, error handling, OpenAPI docs
- Background jobs and queues with BullMQ

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Module 5: Security & Authentication

Duration	Week 12 – 13
Outcome	Applied knowledge of securing enterprise applications against common vulnerabilities with production-grade auth flows.

Topics Covered

- Authentication strategies: session-based, JWT, and OAuth 2.0 / OIDC
- Implementing Auth with NextAuth.js / Auth.js and role-based access control (RBAC)
- Password security: hashing with bcrypt/argon2, salting, and credential storage
- OWASP Top 10: XSS, CSRF, SQL/NoSQL injection, and mitigation strategies
- API security: rate limiting, input validation (Zod), and secure headers (Helmet)
- Secrets management and environment configuration best practices
- Data encryption at rest and in transit; HTTPS/TLS fundamentals
- Compliance basics relevant to enterprise apps: GDPR-aware data handling

Module 6: DevOps, Docker & AWS

Duration	Week 14 – 16
Outcome	End-to-end capability to containerize, deploy, and operate full-stack applications on AWS with CI/CD automation.

Topics Covered

- Docker fundamentals: images, containers, multi-stage builds, and Docker Compose
- CI/CD pipelines with GitHub Actions: automated testing, linting, and builds
- AWS core services: EC2, S3, RDS (PostgreSQL), and IAM roles/policies
- Deploying Next.js apps: AWS Amplify, ECS/Fargate, and Vercel alternatives
- Container orchestration basics with ECS and load balancing (ALB)
- Infrastructure as Code introduction with Terraform / AWS CDK
- Monitoring & logging: CloudWatch, structured logging, and alerting
- Scalability patterns: auto-scaling groups, CDN (CloudFront), and caching strategy
- Capstone: deploying a full MERN/Next.js enterprise app end-to-end on AWS

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Assessment & Certification

- Weekly hands-on coding assignments and code reviews aligned to each module
- Mid-program project: full-stack CRUD application with authentication (Modules 1–5)
- Capstone project: enterprise-grade MERN/Next.js application deployed on AWS with CI/CD
- Final proficiency evaluation covering architecture decisions, security, and deployment

Upon successful completion, participants will demonstrate production-level engineering proficiency across the full modern web stack — from frontend architecture through secure backend systems to cloud deployment.

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